

CS5340 - Uncertainty Modelling in AI

Lecture 2: Bayesian Networks (Directed Graphical Models)

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1 Lecture Overview

- **Motivation:** full joint probability distributions become **intractable** when all variables are (fully) correlated — we need a compact representation.
- **Goal:** introduce **Bayesian Networks (BNs)**, a.k.a. **Directed Graphical Models (DGMs)**, as economical representations of **conditional independence**.
- **Roadmap:**
 1. Define conditional independence and the Markov assumption.
 2. Represent conditional independence in a joint distribution using a Bayesian Network (DAG).
 3. Understand **d-separation** via the three canonical 3-node graphs.
 4. Deduce all conditional independencies in a BN using the **Bayes Ball algorithm**.
 5. Understand the **Markov Blanket** of a node.

Learning Outcomes

By the end of this lecture, you should be able to:

1. Explain the concept of conditional independence.
2. Use a Bayesian Network to represent conditional independence in joint distributions.
3. Describe d-separation using the three canonical 3-node graphs.
4. Deduce all conditional independence in a Bayesian Network using the Bayes Ball algorithm.
5. Explain the concept of the Markov Blanket.

2 Motivation: Why Graphical Models?

It is difficult to work with joint probabilities $p(x_1, \dots, x_N \mid \theta)$ over **fully correlated** random variables.

- Consider N **discrete** random variables $X_1, \dots, X_N$, each $x_i \in \{1, \dots, K\}$.
- Representing the full joint $p(x_1, \dots, x_N \mid \theta)$ requires $O(K^N)$ parameters θ — **inference becomes intractable** for large N , and a huge amount of data is needed to learn all parameters.

Easy Solution? Full Independence

Assuming **full independence** of all variables reduces the parameter count to $O(NK)$:

$$p(x_1, \dots, x_N \mid \theta) = \prod_{i=1}^N p(x_i \mid \theta_i).$$

But is this **always correct**? Consider predicting the next letter in “probabili_” — clearly *not* independent of the letters already known. Or: is a pixel’s label (“water”/“sky”) independent of its neighboring pixels’ labels? **No** — real-world random variables are rarely fully independent, yet fully correlated joint distributions are intractable.

Key Rule: The Compromise — Conditional Independence

A good compromise is an **intermediate degree of dependency**. We want to: (1) compactly represent the joint $p(x_1, \dots, x_N \mid \theta)$ of correlated variables, (2) use it to *infer* one set of variables given another in reasonable time, and (3) *learn* its parameters with a reasonable amount of data. **Use Graphical Models!**

3 Conditional Independence

Definition : Conditional Independence

Two random variables X_A and X_C are **conditionally independent** given X_B if:

$$p(x_A, x_C \mid x_B) = p(x_A \mid x_B) p(x_C \mid x_B),$$

or, equivalently:

$$p(x_A \mid x_B, x_C) = p(x_A \mid x_B), \quad \forall x_B : p(x_B) > 0.$$

That is, learning the value of X_C does **not change** the prediction of X_A once we already know the value of X_B . Written as $X_A \perp X_C \mid X_B$.

Worked Example: Family Trees (Pedigree)

Each node represents an individual’s genotype. **Markov Assumption:** any random variable is locally dependent only on its parent nodes. For Bart, whose parents are Homer and Marge:

$$X_{Bart} \perp (X_{nonDesc} \setminus X_{Parent}) \mid X_{Parent}.$$

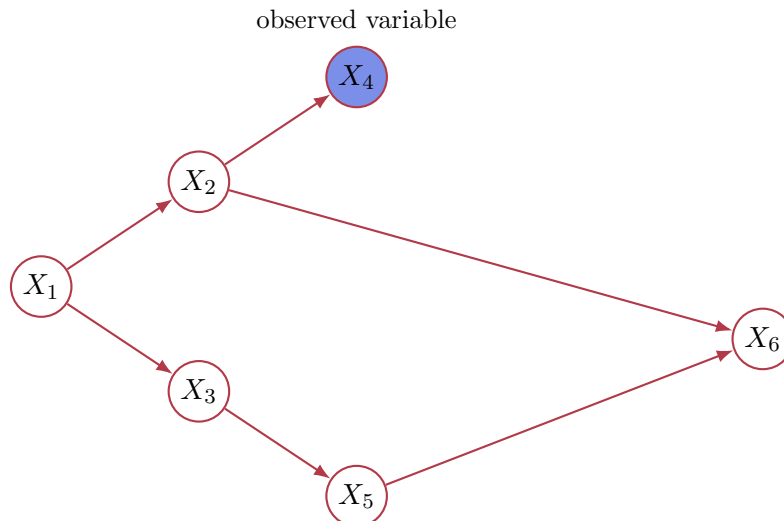
That is, a child’s genotype is conditionally independent of all non-descendants (e.g. aunts, grandparents) given its parents’ genotypes.

4 Bayesian Networks: Definitions

Definition : Bayesian Network / Directed Graphical Model

We use a **Directed Acyclic Graph (DAG)** $\mathcal{G}(\mathcal{V}, \mathcal{E})$ to represent conditional independence, where $\mathcal{V}$ is a set of **nodes** (random variables) and $\mathcal{E}$ a set of **oriented edges** (direct dependencies). $\mathcal{G}$ contains **no cycles**.

- A **shaded** node refers to an **observed** variable; an **unshaded** node refers to a **latent/hidden** (unobserved) variable.
- Each node $i \in \mathcal{V}$ has an associated random variable X_i and a set of **parent nodes** π_i (possibly empty). Let X_{π_i} denote all random variables that are parents of X_i .



- **Ancestors** $\text{anc}(t) \triangleq \{s : s \rightsquigarrow t\}$ — parents, grandparents, etc., that connect to t via a trail. E.g. $\text{anc}(X_4) = \{X_1, X_2\}$.
- **Descendants** $\text{desc}(s) \triangleq \{t : s \rightsquigarrow t\}$ — children, grandchildren, etc., reachable via trails from s . E.g. $\text{desc}(X_3) = \{X_5, X_6\}$.
- For the graph above: $X_{\pi_1} = \emptyset$, $X_{\pi_2} = X_1$, $X_{\pi_3} = X_1$, $X_{\pi_4} = X_2$, $X_{\pi_5} = X_3$, $X_{\pi_6} = \{X_2, X_5\}$.

5 The Markov Assumption

Key Rule: Markov Assumption

Each random variable X_i is independent of its **non-descendants** $X_{\text{nonDesc}(X_i)}$ given its **parents** X_{π_i} :

$$\{X_i \perp (X_{\text{nonDesc}(X_i)} \setminus X_{\pi_i}) \mid X_{\pi_i}\}.$$

Worked Example: Basic Conditional Independencies from the DAG

For the 6-node DAG above, the Markov assumption gives:

$$\begin{array}{ll} X_1 \perp \emptyset \mid \emptyset, & X_4 \perp \{X_1, X_3, X_5, X_6\} \mid X_2, \\ X_2 \perp \{X_3, X_5\} \mid X_1, & X_5 \perp \{X_1, X_2, X_4\} \mid X_3, \\ X_3 \perp \{X_2, X_4\} \mid X_1, & X_6 \perp \{X_1, X_3, X_4\} \mid \{X_2, X_5\}. \end{array}$$

6 Bayesian Networks: Joint Probability Factorization

Key Rule: Factorization of the Joint Distribution

The basic conditional independence statements of the DGM give rise to a set of **local conditional probabilities** $p(x_i | x_{\pi_i})$, $i = 1, \dots, N$, each defined locally according to the parent-child relationship specified by the DGM. The **joint probability** can be read off the graph as the product of all local conditional probabilities:

$$p(x_1, \dots, x_N) = \prod_{i=1}^N p(x_i | x_{\pi_i})$$

Proof Sketch

By the **chain rule** (assuming a topological ordering of the DGM):

$$p(x_1, \dots, x_N) = p(x_1) \prod_{i=2}^N p(x_i | x_{x_1, \dots, x_{i-1}})$$

$$\xrightarrow{\{X_i \perp (X_{\text{nonDesc}(x_i)} \setminus X_{\pi_i}) | X_{\pi_i}\}} p(x_1, \dots, x_N) = \prod_{i=1}^N p(x_i | x_{\pi_i}).$$

Worked Example: Reading Off the Joint from the DAG

For the 6-node DAG above:

$$p(x_1, \dots, x_6) = p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2) p(x_5 | x_3) p(x_6 | x_2, x_5).$$

Verifying $X_4 \perp \{X_1, X_3\} | X_2$, i.e. $p(x_4 | x_1, x_2, x_3) = p(x_4 | x_2)$: first marginalize to get $p(x_1, x_2, x_3, x_4)$ and $p(x_1, x_2, x_3)$,

$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= \sum_{x_5} \sum_{x_6} p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2) p(x_5 | x_3) p(x_6 | x_2, x_5) \\ &= p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2) \underbrace{\sum_{x_5} p(x_5 | x_3) \sum_{x_6} p(x_6 | x_2, x_5)}_{=1} \\ &= p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2), \\ p(x_1, x_2, x_3) &= \sum_{x_4} p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2) = p(x_1) p(x_2 | x_1) p(x_3 | x_1). \end{aligned}$$

Dividing the two marginals gives $p(x_4 | x_1, x_2, x_3) = p(x_4 | x_2)$, confirming $X_4 \perp \{X_1, X_3\} | X_2$. ■

7 Parameter Reduction**Key Rule: Parameter Count**

Let m_i denote the number of parents of node X_i , and each node takes K values. The conditional probability table for X_i has size K^{m_i+1} . This gives a **huge reduction** in the number of parameters needed to represent the joint — from $O(K^N)$ (fully dependent) to $O(K^{m+1})$ where $m = \max_i m_i \ll N$ (using the BN factorization).

Worked Example: Binary Random Variables

For the 6-node DAG with binary $x_i \in \{0, 1\}$: X_1 needs 2 params (no parents); X_2, X_3, X_5 each need 4 params (1 parent each: $K^{1+1} = 4$); X_4 needs 4 params (parent X_2); X_6 needs 8 params (2 parents $\{X_2, X_5\}$: $K^{2+1} = 8$). **Total** = $2 + 4 + 4 + 4 + 4 + 8 = 26$ parameters, versus $2^6 = 64$ for the fully dependent joint — and the gap grows dramatically with more nodes.

8 Beyond the Markov Assumption: Motivating d-Separation

It turns out that $X_1 \perp X_6 \mid \{X_2, X_3\}$ also holds, even though this is **not** directly observed from the parent-child (Markov) relations above.

Proof

$$p(x_1, x_2, x_3, x_6) = p(x_1)p(x_2|x_1)p(x_3|x_1) \sum_{x_5} p(x_5|x_3)p(x_6|x_2, x_5),$$

$$p(x_2, x_3, x_6) = \sum_{x_1} p(x_1)p(x_2|x_1)p(x_3|x_1) \sum_{x_5} p(x_5|x_3)p(x_6|x_2, x_5).$$

Dividing: $p(x_1 \mid x_2, x_3, x_6) = \frac{p(x_1, x_2, x_3, x_6)}{p(x_2, x_3, x_6)} = \frac{p(x_1, x_2, x_3)}{p(x_2, x_3)} = p(x_1 \mid x_2, x_3)$, confirming $X_1 \perp X_6 \mid \{X_2, X_3\}$.

Question: Can we deduce all such conditional independencies by just **inspecting the DAG**, without complicated algebra?

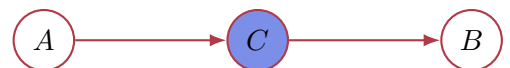
Answer: Yes — observe that the nodes $\{X_2, X_3\}$ “**block**” all paths from X_1 to X_6 . This suggests the notion of **graph separation**. We must, however, be careful: e.g. X_2 is **not** independent of X_3 given $\{X_1, X_6\}$, even though a naïve reading of “blocking” might suggest otherwise. The precise definition of blocking requires the **three canonical 3-node graphs** and **d-separation**.

9 The Three Canonical 3-Node Graphs

9.1 1. Head-to-Tail



$$A \not\perp B \mid \emptyset$$



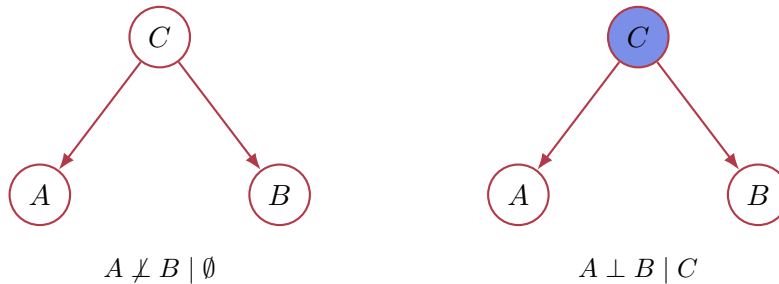
$$A \perp B \mid C$$

Joint: $p(a, b, c) = p(a)p(c \mid a)p(b \mid c)$. If **none** observed, marginalizing over C gives $p(a, b) = p(a) \sum_c p(c \mid a)p(b \mid c) = p(a)p(b \mid a) \neq p(a)p(b)$ in general, so $A \not\perp B \mid \emptyset$.

If we **condition on** C : $p(a, b \mid c) = \frac{p(a, b, c)}{p(c)} = \frac{p(a)p(c \mid a)p(b \mid c)}{p(c)} \stackrel{\text{Bayes}}{=} p(a \mid c)p(b \mid c)$, giving $A \perp B \mid C$.

Node C is **head-to-tail** w.r.t. the path $A \rightarrow C \rightarrow B$: such a path connects A, B and renders them dependent, but **observing** C blocks the path. *Interpretation:* Past $\rightarrow$ Present $\rightarrow$ Future — “the past is independent of the future given the present” (a simple classical **Markov Chain**).

9.2 2. Tail-to-Tail



Joint: $p(a, b, c) = p(a \mid c)p(b \mid c)p(c)$. If none observed, $p(a, b) = \sum_c p(a \mid c)p(b \mid c)p(c)$ does **not** factorize into $p(a)p(b)$, so $A \not\perp B \mid \emptyset$. Conditioning on C : $p(a, b \mid c) = p(a, b, c)/p(c) = p(a \mid c)p(b \mid c)$, giving $A \perp B \mid C$.

Node C is **tail-to-tail** w.r.t. the path from A to B ; observing C blocks the path. *Interpretation:* C = “chronological age”, A = “shoe size”, B = “gray hair”. Given a person’s age, there is no further relationship between shoe size and gray hair — C “**explains**” the observed dependence between A and B .

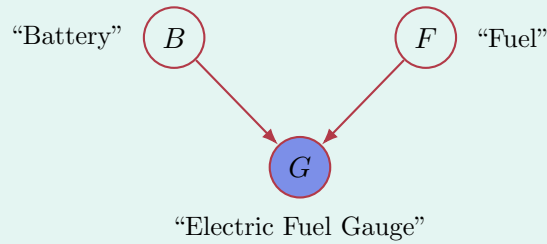
9.3 3. Head-to-Head (V-structure)



Joint: $p(a, b, c) = p(a)p(b)p(c \mid a, b)$. If none observed, marginalizing over C : $p(a, b) = \sum_c p(a)p(b)p(c \mid a, b) = p(a)p(b)$, so $A \perp B \mid \emptyset$! Conditioning on C : $p(a, b \mid c) = p(a)p(b)p(c \mid a, b)/p(c)$, which **does not** factorize in general, so $A \not\perp B \mid C$.

Node C is **head-to-head** (a “**v-structure**”) w.r.t. the path $A \rightarrow C \leftarrow B$. When C is **unobserved**, it **blocks** the path and A, B are independent; but conditioning on C (or any of its descendants) **unblocks** the path, rendering A, B dependent — this is the “**explaining-away**” effect.

Worked Example: Explaining Away — Battery, Fuel, and the Gauge



$B \in \{0, 1\}$ (flat/charged), $F \in \{0, 1\}$ (empty/full), $G \in \{0, 1\}$ (empty/full reading). Given:

$$p(B=1) = 0.9, \quad p(F=1) = 0.9, \quad p(G=1|B, F) : \begin{cases} (1, 1) \rightarrow 0.8 \\ (1, 0) \rightarrow 0.2 \\ (0, 1) \rightarrow 0.2 \\ (0, 0) \rightarrow 0.1 \end{cases}$$

Prior: $p(F=0) = 0.1$.

Observe $G=0$ (gauge reads empty):

$$p(G=0) = \sum_{b,f} p(G=0|b, f)p(b)p(f) = 0.315, \quad p(G=0|F=0) = \sum_b p(G=0|b, F=0)p(b) = 0.81,$$

$$p(F=0 | G=0) = \frac{p(G=0|F=0)p(F=0)}{p(G=0)} \approx \mathbf{0.257} > p(F=0) = 0.1.$$

Observing the gauge reads empty makes it **more likely** the tank is empty, as expected.

Now also observe $B=0$ (battery flat):

$$p(F=0 | G=0, B=0) = \frac{p(G=0|B=0, F=0)p(F=0)}{\sum_f p(G=0|B=0, f)p(f)} \approx \mathbf{0.111} < 0.257.$$

Finding the battery is flat **“explains away”** the observation that the gauge reads empty — B and F have become dependent given G ! This illustrates the head-to-head structure.

Unblocking via Descendants

More generally, the observation of **any descendant** of a head-to-head node C (not just C itself) also unblocks the path from A to B .

10 Graph Separation and d-Separation

Definition : d-Separation

$A \perp B \mid C$ holds if all paths from nodes in set A to nodes in set B are **“blocked”** when all nodes in set C are observed. A is said to be **d-separated** from B by C , and the joint distribution over all variables in the graph will satisfy $A \perp B \mid C$.

Key Rule: Blocking / d-Separation Rule

From the three canonical 3-node graphs, a path is “blocked” / d-separated if it includes a node such that **either**:

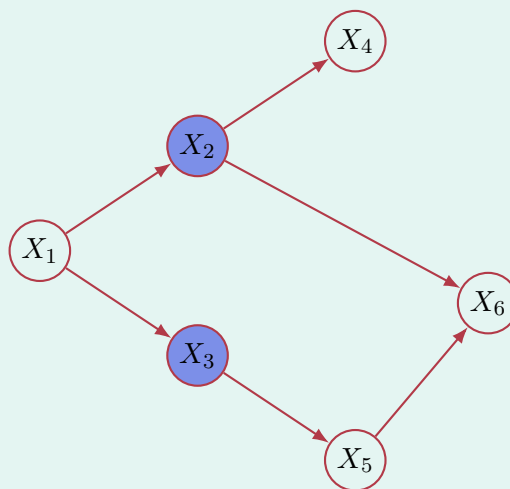
- (a) the arrows on the path meet either **head-to-tail** or **tail-to-tail** at the node, and the node **is** in set C ; **or**
- (b) the arrows meet **head-to-head** at the node, and **neither** the node **nor any of its descendants** is in set C .

11 The Bayes Ball Algorithm

Algorithm: Bayes Ball — a Reachability Algorithm

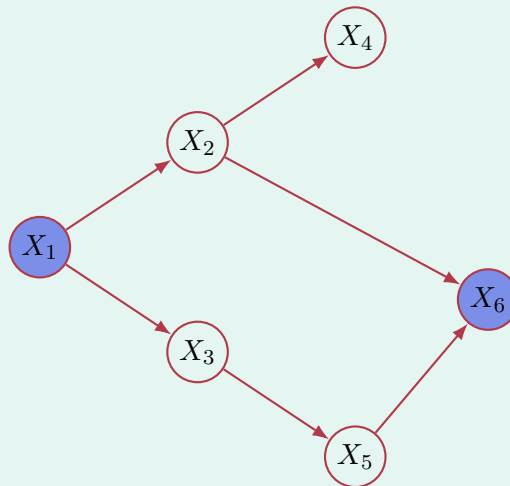
Input: Graph $\mathcal{G}$, source set A , target set B , observed set C .

1. Shade the nodes in set C .
 2. Place a ball at each of the nodes in set A .
 3. Let balls bounce around the graph according to the d-separation rules:
 - **Head-to-tail / tail-to-tail:** if the current node is observed, **stop**; else pass the ball to its children/onward.
 - **Head-to-head:** if the node or any of its descendants is observed, pass the ball **to its parents**; else **stop**.
 4. **If** any ball reaches a node in B , conclude $A \not\perp B \mid C$; **else** $A \perp B \mid C$.
- Can be implemented as a **breadth-first search**.

Worked Example: 1 — Blocked Path

With $C = \{X_2, X_3\}$ observed, a ball starting at X_1 arrives head-to-tail at X_2 (observed) and at X_3 (observed) — both **stop** the ball. It **cannot pass** through X_2 to X_6 , nor through X_3 : hence $X_1 \perp X_6 \mid \{X_2, X_3\}$.

Worked Example: 2 — Unblocked Path via a Head-to-Head Descendant

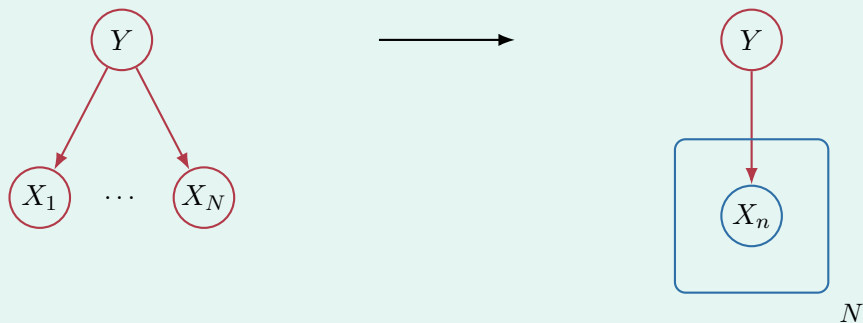


With $C = \{X_1, X_6\}$ observed, starting at X_2 : the path $X_2 \rightarrow X_6 \leftarrow X_5$ meets **head-to-head** at X_6 , which **is** observed — so the ball **passes through** X_6 to its parent X_5 , then on to X_3 . The ball **can pass** through $X_2 \rightarrow X_6 \rightarrow X_5 \rightarrow X_3$: hence $X_2 \not\perp X_3 \mid \{X_1, X_6\}$.

Worked Example: 3 — Naïve Bayes Classifier

Task: classify e-mails as spam ($Y=1$) or not spam ($Y=0$). Let $1:N$ index a vocabulary; $X_n = 0$ if word n appears in the e-mail, 1 otherwise. E-mails are drawn from $p(x_1, \dots, x_N, Y)$.

Plate notation: N nodes,
only X_n shown explicitly



Joint distribution (using the head-to-head / v-structure factorization, since Y is the common parent):

$$p(x_1, \dots, x_N, Y) = p(x_1, \dots, x_N \mid Y) p(Y) = \prod_{n=1}^N p(x_n \mid Y) p(Y)$$

— each X_n is conditionally independent of the others given Y (“**naïve**” assumption).

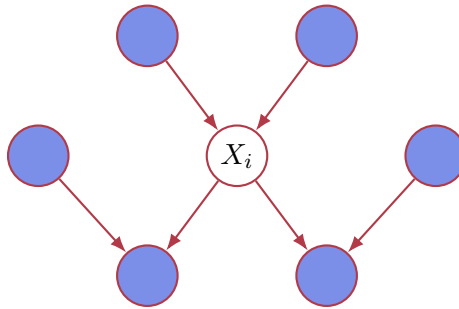
Prediction via Bayes’ rule:

$$p(Y=1 \mid x_1, \dots, x_N) = \frac{p(Y=1) \prod_{n=1}^N p(x_n \mid Y=1)}{\sum_{y \in \{0,1\}} p(Y=y) \prod_{n=1}^N p(x_n \mid Y=y)}.$$

12 The Markov Blanket

Definition : Markov Blanket

The **Markov blanket** of a node X_i comprises the set of its **parents**, **children**, and **co-parents** (the other parents of its children).

**Key Rule: Markov Blanket Property**

The conditional distribution of X_i , conditioned on **all remaining** variables in the graph, depends **only** on the variables in its Markov blanket:

$$p(x_i | x_{\{j \neq i\}}) = \frac{\prod_k p(x_k | x_{\pi_k})}{\int \prod_k p(x_k | x_{\pi_k}) dx_i} = \frac{\underbrace{\prod_m p(x_m | x_{\pi_m} : x_i \in x_{\pi_m})}_{\text{children \& co-parents of } X_i} \underbrace{p(x_i | x_{\pi_i})}_{\text{parents of } X_i}}{\int \prod_m p(x_m | x_{\pi_m} : x_i \in x_{\pi_m}) p(x_i | x_{\pi_i}) dx_i}$$

(all terms not involving x_i cancel between numerator and denominator). This property is **useful for inference** and **Gibbs sampling** (see Lecture 9).

13 Applications

- **Urban scene understanding:** a Bayesian network jointly models road parameters $\mathcal{R}$, vehicle tracklets, vanishing points, scene labels, scene flow, and occupancy grids, factorizing as a product of local conditionals over these variables (Geiger, PhD Thesis 2013).
- **Robust 3D point cloud reconstruction:** odometry constraints $X_{i,i+1}$ and loop-closure constraints Y_{ij} are modeled with assignment variables Z_{ij} in a Bayesian network, solved via Expectation-Maximization (EM): E-step $Q_{EM} = \sum_z p(Z | Y, T^{old}) \ln p(X, Y, Z | T)$; M-step $\arg \max_T Q_{EM}$ (Lan, Yew, Lee, *CVPR 2019*).

14 Summary

Key Takeaways

1. **Conditional independence** $X_A \perp X_C \mid X_B$ lets us model an *intermediate* degree of dependency between full independence and full correlation.
2. **Bayesian Networks** (DAGs) compactly encode conditional independence via the **Markov assumption**, factorizing the joint as $p(x_1, \dots, x_N) = \prod_i p(x_i \mid x_{\pi_i})$ — reducing parameters from $O(K^N)$ to $O(K^{m+1})$.
3. The **three canonical 3-node graphs** — head-to-tail, tail-to-tail, head-to-head — characterize how paths are blocked or unblocked, leading to **d-separation**.
4. The **Bayes Ball algorithm** deduces *all* conditional independencies in a BN by simulating balls bouncing along blocked/unblocked paths (a BFS-style reachability test).
5. The **Markov Blanket** (parents, children, co-parents) is the minimal set of variables that renders a node conditionally independent of everything else — key for inference and Gibbs sampling.
6. These tools are the foundation for efficient probabilistic inference in AI, with applications spanning scene understanding to robust 3D reconstruction.

Acknowledgement: Slide content adapted from M. I. Jordan, “An Introduction to Probabilistic Graphical Models” (2002), Section 2.1; C. Bishop, “Pattern Recognition and Machine Learning”, Ch. 8; K. Murphy, “Machine Learning: A Probabilistic Approach”, Ch. 10; D. Koller & N. Friedman, “Probabilistic Graphical Models”, Ch. 3.